

Research Paper

Family dynamics on data tracking and sharing with child-oriented wearables

Isil Oygur^{a,*}, Yunan Chen^b, Daniel A. Epstein^b^a Ullman School of Design, University of Cincinnati, Cincinnati, OH, USA^b Donald Bren School of Information and Computer Sciences, University of California, Irvine, Irvine, CA, USA

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ABSTRACT

Child-oriented wearables, such as activity trackers, smartwatches, and GPS devices, are increasingly framed as tools for safety, health, and connectivity. Beyond individual self-tracking, these devices also serve as expressions of parental care through shared data practices. This paper examines how these technologies mediate children's understanding of datafication and dataveillance, and why families adopt them by examining the benefits and values families attribute to their use. We report on 12 co-design sessions, each involving a parent and at least one child. Our findings support prior research on the early normalization of datafication and dataveillance without a corresponding depth of understanding of these concepts and processes. These technologies do not automatically foster awareness of data practices. Normalization of data tracking and sharing is reinforced both by the introduction of wearables as expressions of parental care and family values and by the lack of sustained child-parent dialogue, which emerges as a critical factor. The co-design sessions reveal both willingness and openness among children and parents to engage in such dialogue, while also suggesting that these conversations require structured facilitation to occur regularly. As situational mediators, wearables function alternately as conversation starters and as sources of tension. We argue that, for these technologies to support reflective mediation, future designs should prioritize family co-use, incorporate prompts that support meaningful dialogue about data tracking and sharing, and make datafication and dataveillance processes transparent to enhance children's understanding and agency.

1. Introduction

Child-oriented wearable technologies, such as activity trackers, smartwatches, and GPS devices specifically designed for children, have become increasingly prevalent in everyday life (Cha et al., 2025; Jørgensen et al., 2016; Oygür et al., 2020; Su & Chen, 2024). Like many other digitally enhanced technologies for children - such as smart toys, robots, and tablets - these devices collect data from children. Indeed, the core premise of this product category is the collection of personal data, which is then transformed into quantified information for purposes of self-awareness, behavior change, communication, and/or parental tracking. However, the data practices they embody are inherently distinct from those of other technologies, necessitating an analysis of them as a separate category. Accordingly, a growing body of literature (Gram-Hansen, 2019; Jørgensen et al., 2016; Saksono et al., 2018; Silva et al., 2024; Zehrung et al., 2021) has begun to examine these technologies specifically.

Child-oriented wearables are characterized by three distinct data practices that differentiate them from other digitally enhanced technologies for children. First, unlike other mobile and internet-connected devices, child-oriented wearables enable pervasive and embodied data collection (Gram-Hansen, 2019; Lupton & Williamson, 2017). Children can wear these devices continuously throughout the day without health reasons, often even while sleeping. As a result, personal data is collected passively and continuously in the background, rather than only during moments of direct interaction.

Second, these technologies introduce a different form of parental tracking compared to other technologies (Bettany & Kerrane, 2016; Kuzminykh & Lank, 2019; Oygür et al., 2020). Although they are primarily marketed as tools to promote children's safety, health, well-being, and communication while supporting children's independence and autonomy, their use often extends beyond self-tracking. Parents frequently adopt these devices as tools for reassurance and monitoring, thereby contributing to evolving forms of parental tracking

* Corresponding author.

E-mail addresses: oyguril@uc.edu (I. Oygur), yunanc@ics.uci.edu (Y. Chen), epstein@ics.uci.edu (D.A. Epstein).<https://doi.org/10.1016/j.ijcci.2026.100848>

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practices (Bettany & Kerrane, 2016; Mackintosh et al., 2019; Wang et al., 2017). However, the nature of this tracking differs from that of conventional parental control applications, which are typically designed to restrict children's online activities (Akter et al., 2022; Ali et al., 2020; Ghosh et al., 2018). In contrast, child-oriented wearables are marketed directly to children, with parents positioned as secondary or silent users.

Third, the significance of these technologies lies in their purpose: quantifying human behavior. While early wearable technologies primarily targeted adults, the introduction of child-oriented wearables has extended these practices to children as young as three years old (Oygür et al., 2020). Children at young ages often do not fully understand the implications of using data-centric devices (Manches & Plowman, 2021; Mascheroni & Siibak, 2021; Sun et al., 2024). As such, these devices serve as embodied representations of broader social and ethical concerns, including the normalization of data tracking and sharing during early childhood (Jørgensen et al., 2016; Lupton & Williamson, 2017; Mascheroni & Siibak, 2021).

Taken together, these three qualities position child-oriented wearables as significant additions to everyday family life, with the potential for long-term impacts on children. Existing research has largely focused on their technical aspects (e.g., Fúster et al., 2024; Saatjohann et al., 2020), their role in managing chronic health conditions (e.g., Kaziunas et al., 2017) and special needs (e.g., Cha et al., 2025), their contribution to promoting health and well-being (e.g., Saksono et al., 2018), and their use in enhancing child safety (e.g., Simpson, 2014). Thus, the existing body of work has primarily focused on the functional affordances of these devices. Studies that examine how these devices mediate datafication and dataveillance (outside the context of chronic health conditions) and contribute to the normalization of these processes are limited. Such an understanding is critical, as early encounters with wearable technologies may shape children's long-term attitudes toward datafication, dataveillance, autonomy, and technology-mediated communication (Lupton & Williamson, 2017; Mascheroni, 2020). To this end, we conducted a co-design study guided by two research questions.

- 1) What benefits and values do children and parents attribute to child-oriented wearables?
- 2) If and how do child-oriented wearables mediate children's understanding of pervasive datafication and dataveillance?

Using various design probes, we conducted 12 co-design sessions, each involving a parent and at least one elementary- or intermediate-school-aged child. We explored both children's and parents' preferences regarding child-oriented wearables, the features they find valuable, and how wearables mediate conversations, expectations, and practices within families. We also examined how different types of data and tracking practices shape family discussions and interactions.

Given our focus on family-level integration, we adopted a socio-technical lens in our analysis, drawing on Peter-Paul Verbeek's concept of technological mediation (Verbeek, 2016, 2005). In response to our research questions, we found that children and parents associated child-oriented wearables with overlapping values of care, safety, and connectivity, and understood data related to these values, such as location data, as expressions of family values and parental care. However, they differed in their expectations about which data should be tracked and with whom more personal data should be shared, and children and parents did not often discuss these issues explicitly. During our co-design sessions, we observed how wearables became situational and episodic mediators, prompting child-parent conversations about data while also exposing tensions concerning privacy, access, and the boundary between personal and family-related information. We discuss how the adoption of child-oriented wearables as expressions of family values and parental care, combined with a lack of sustained dialogue, contributes to children's normalization of data tracking and sharing without a full understanding of the underlying data practices.

Our findings contribute to the growing literature on child-oriented wearables, parental tracking, and technology use in family life in three ways.

- Children distinguish between data they perceive as personal and data they regard as family-related. Although child-oriented wearables reflect children's and parents' overlapping values of care through location tracking and continuous connectivity, children establish different expectations and boundaries around data they consider more personal.
- Children's normalization of datafication and dataveillance is shaped not only by the use of child-oriented wearables, but also by family dynamics surrounding their adoption. When these technologies are introduced as expressions of family values and parental care without explicit and sustained conversations about data, children may come to accept datafication and dataveillance from an early age without a deep understanding of how these processes operate.
- As technological mediators, child-oriented wearables function differently at different levels. Although they continuously contribute to the normalization of datafication and dataveillance through routine use, their contribution to children's reflection on and understanding of these processes is situational and episodic. Current designs do not systematically support sustained child-parent dialogue or children's critical reflection on data practices.

2. Related work

Our study builds upon previous work spanning several intersecting areas. These include self-tracking, family informatics, family-centered design, technology use in family life, and datafication of childhood. We reviewed these areas of the literature through a socio-technical lens, examining the co-use practice of child-oriented wearables and how these technologies mediate children's understanding of datafication and dataveillance.

2.1. Children and the datafication of their lives with wearables

Personal tracking and data-driven technologies were initially designed with adult users in mind, but in recent years, they have increasingly become part of children's everyday lives (Oygür et al., 2021; Zehrung et al., 2021). Today, children are exposed to a growing range of wearable technologies, including activity trackers, smart-watches, and GPS trackers, that are explicitly targeted to younger users. Such child-oriented wearables have experienced steady market growth since their introduction and are projected to continue growing (Global Market Insights, 2025; Patel et al., 2024). This expansion has meant that many children are introduced to pervasive datafication and dataveillance practices from an early age (Lupton & Williamson, 2017). Often, they engage with data-centric technologies even before they are aware of what data is (Sun et al., 2024) or the implications of continuous data tracking (Mascheroni, 2020).

Children's wearables serve a variety of functions. In the health domain, they are deployed to manage chronic conditions and special needs, such as asthma (van der Kamp et al., 2020), diabetes (Shin and Holtz, 2020), autism spectrum disorder (Koumpouros & Toulas, 2020), and ADHD (Cibrian et al., 2020). Beyond health needs, families adopt these technologies for more everyday purposes, including supporting communication and social interaction (Liu et al., 2024), monitoring location and safety (Hasinoff, 2017), and tracking health and well-being metrics such as exercise (Saksono et al., 2015, 2020), healthy eating (Schaeffbauer et al., 2015), and sleep (Pina et al., 2020). These uses underscore the broadening scope of wearable technologies from medical management to lifestyle support.

Although initially modeled after adult self-tracking solutions, child-oriented wearables require a fundamentally different understanding of use practices. For younger children in particular, wearables are rarely

operated in isolation. Parents and caregivers typically play an active role (Oygur Ilhan et al., 2024), either by accessing their child's health and behavioral data through companion apps (Mackintosh et al., 2019), co-reflecting on data with children (Lee et al., 2024), or by helping children use wearables (Oygür et al., 2020). Wearables have become part of the digital parenting tools of contemporary Western societies (Livingstone & Blum-Ross, 2020). Parents see these technologies as a way to express their care for their children (Lupton, 2020). This recognition of relational care in households and families has given rise to the notion of family informatics, an area of research in which technologies mediate care through collaborative data-tracking practices among family members (Pina et al., 2017). Within family informatics, child-parent tracking practices highlight distinct dynamics, since parents carry primary responsibilities and children depend on their parents for health- and well-being-related issues (Pina et al., 2017).

From one perspective, wearables promise to foster independence, autonomy, and self-regulation among children. Parents often turn to these devices to encourage children to take responsibility, using wearable feedback to track progress toward health and behavioral goals, as well as chore accomplishments (Gram-Hansen, 2019). The devices also enable parents to maintain ubiquitous access to their children via calls, texts, and location tracking, allowing children to navigate independently with a greater sense of safety while parents gain peace of mind (Murphy & Wu, 2025). Children themselves often report positive experiences. They feel “grown up” by owning smart technologies similar to those used by adults (Oygür et al., 2020). They enjoy gamified behavior-change features and technology-mediated social interaction (Oygür et al., 2021), and they may experience an enhanced sense of safety and empowerment through self-tracking (Ervasti et al., 2016; Sukk & Siibak, 2021).

Yet, these same practices can also generate unintended negative consequences. The collaborative use of wearables can produce tensions within families, highlighting dichotomies such as parental involvement versus children's independence, or co-regulation versus autonomy (Czeskis et al., 2010; Silva et al., 2023). While some conversations around tracked data foster collaboration and mutual learning, others become sources of conflict (Cha et al., 2025) and surveillance (Mavoa et al., 2023). Scholars have also shared increased parental worry and anxiety through tracking behaviors (Kuzminykh & Lank, 2019). Moreover, most commercial devices afford children little agency in deciding how their data is collected, shared, or interpreted. This lack of control becomes more salient as children grow older. For instance, Thompson et al. (2024) found that primary school children were more willing to share data than secondary school children, who increasingly viewed parental monitoring as intrusive. In another study of teens with an average age of 15, Czeskis et al. (2010) reported how their participants differentiated between internal and external information. While teens were comfortable sharing external information, such as their location, they did not want their internal states, such as their mood, to be known to their parents. In such cases, wearables risk being perceived as threats to privacy, autonomy, and trust (Hasinoff, 2017; Mancini et al., 2011; Simpson, 2014).

Scholars also warn against the possible normalization of datafication and dataveillance at an early age through the use of wearables in everyday life (Lupton & Williamson, 2017; Mascheroni & Siibak, 2021). In their study with seven- to 12-year-old children, Sun et al. (2024) reported a lack of conceptual understanding of data among children. Their “participants conceptualized data collection concretely, lacked awareness of the pervasive nature of data ...” (Sun et al., 2024, p. 518). Given this, while parents might view child-oriented wearables as empowering, children's use of these technologies, as well as parental tracking practices, may inadvertently socialize children into accepting datafication and dataveillance as a default condition of everyday life (Hasinoff, 2017).

Existing studies have examined what children and parents think about and do with tracked data (Jørgensen et al., 2016; Oygür et al.,

2021; Pina et al., 2020; Saksono et al., 2020). However, they provide less insight into the family dynamics through which datafication and dataveillance become normalized, including how children reason about pervasive data collection and sharing, negotiate boundaries around parental access, and weigh safety against privacy and self-regulation. We address this gap by examining children and parents together and by bringing research on self-tracking, family informatics, family-centered design, family technology use, and the datafication of childhood into conversation. This approach enables us to investigate how family values, parental care, and child-parent dialogue shape children's understanding and normalization of wearable data practices.

2.2. Family dynamics and the mediation of children's technology use

The integration of new technology and media into family life has been widely studied, with research pointing to their ambivalent effects on children (Coyne et al., 2013), family dynamics, and interpersonal relationships, including child-parent interaction (Shin et al., 2021). Technologies such as conversational agents, social robots, tablets, and smartphones can enrich children's learning, social engagement, interactional routines, and self-management (Cagiltay et al., 2020). At the same time, their use causes conflicts between children and parents, often stemming from value tensions around autonomy, privacy, and control (Blackwell et al., 2016; Livingstone, 2002; Nouwen et al., 2017). How technologies are introduced into family life, the reasons for their use, children's age, and the parenting styles that shape mediation strategies all influence whether their consequences are experienced as positive or negative.

Parents adopt different strategies around their children's engagement with technology and online safety (Livingstone & Helsper, 2008; Clark, 2011). These strategies range from restrictive approaches that limit access and use to instructive or co-engaged approaches that actively guide their children's technology use and reflect on content and practices with them. Some parents are turning child-technology interactions into joint media engagements to teach digital literacy, online safety, and privacy (Takeuchi & Stevens, 2011), while others postpone conversations about privacy and online safety (Kumar et al., 2017). There are also parents who adopt little or no mediation or involvement at all. These parents may want their children to practice responsibility and self-regulation, or they may wish to signal trust, particularly as children approach adolescence and adulthood (Vaterlaus et al., 2015). Recent studies highlight the effectiveness of collaboration (Nouwen et al., 2017) and co-management (Akter et al., 2022) over parental mediation for raising resilient children and overcoming child-parent conflicts on technology and media use (Park et al., 2024). Through their parenting style, reflected in strategies for navigating child-technology interactions, parents play a significant role in their child's digital upbringing (Barron et al., 2009).

From this perspective, parents are instrumental in shaping their child's initial encounters with technology (Jennings, 2020). Parental values are translated to family values through the use and non-use of technology. For example, Vasalou et al. (2012) found that some UK parents deliberately chose not to use location-tracking technologies, interpreting them as incompatible with values of trust and independence. In contrast, Murphy & Wu (2025) describe how parents who migrated from rural areas to central China regard tracking behaviors as an essential aspect of childcare.

Depending on patterns of use and non-use, family conversations, parental strategies, and the child's developmental stage, children can develop different perceptions of a technology. These dynamics underscore that technologies in family life are not neutral; they both reflect and shape the moral economies of trust, independence, and accountability within households. This perspective aligns with Peter-Paul Verbeek's theory of technological mediation (Verbeek, 2006, 2016).

Verbeek (2006) conceptualizes technological mediation as a useful theory “for analyzing the role of technologies in the daily lives of human

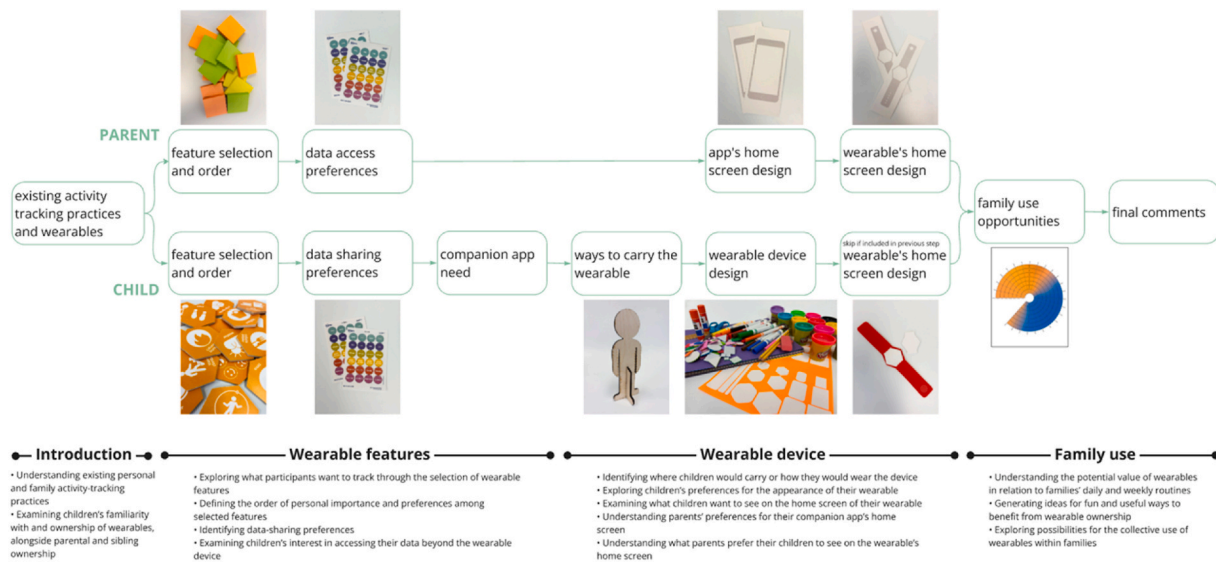


Fig. 1. Phases of the co-design protocol, the probes used during the sessions, and the aims of each phase.

beings" (p. 363). Building on the work of Don Ihde and Science and Technology Studies (STS), particularly Actor-Network Theory, he argues that technologies mediate human-world relations by shaping how reality is perceived, interpreted, and acted upon (Verbeek, 2005, 2016). Accordingly, mediation is context-dependent, and studying technology requires examining human actions and experiences alongside technological artifacts. Verbeek (2016) emphasizes that understanding mediation involves not only analyzing "what things do" but also "how humans give meaning to these mediations - both empirically and conceptually" (p. 3). This perspective distinguishes his approach from broader STS accounts of mediation by giving greater attention to how humans experience and interpret the ways technologies shape their actions. His framework also foregrounds the ethical dimension of design, suggesting that technologies materialize moral values as they mediate human-world relations (Verbeek, 2006).

From our perspective, this framework provides a useful lens for studying child-oriented wearables for two main reasons. First, these technologies inherently rely on human interpretation of data, shaping how users experience and make sense of the world. Second, the framework enables a more holistic understanding of human-technology entanglements, shifting the focus from parents' role, mediation, and co-use to the broader role of technologies in structuring human-world relations. This shift enables a more nuanced analysis of how family dynamics are shaped by the introduction and everyday use of these technologies, and vice versa. This is also what we aimed for in our study. We used the lens of technological mediation to examine how child-oriented wearables contribute to the normalization of these processes and mediate datafication and dataveillance.

3. Methods

We adopted a co-design approach to explore our research questions. Co-design has been widely used to explore existing and emerging technologies with children (Cha et al., 2025; Yip et al., 2016) because it supports children's agency, encourages creative expression, and empowers them as active agents in the design and development of technology. Each session included a parent and at least one neurotypical child. Conducting joint sessions was strategically selected to gain deeper insights into potential child-parent interactions mediated by wearables.

3.1. Co-design materials and protocol

Our co-design protocol consisted of four main phases: 1. Introduction, 2. Wearable features, 3. Wearable device, and 4. Family use (see Fig. 1). All sessions were conducted by the first author, who had prior experience collecting data with children and facilitating joint sessions with children and parents. While we acknowledge that parental power dynamics cannot be fully eliminated, we took measures to support children's agency and expression. To mitigate potential power imbalances and encourage open participation from both parties, we structured a co-design protocol that included both individual and joint activities. Hands-on activities enabled children to express their ideas non-verbally and engage on their own terms. During joint activities, the researcher actively encouraged children's participation by directing questions to them first, creating space for their responses before inviting parental input.

At the beginning of each session, the details of the co-design activities were explained verbally, and both children and parents were given time to ask questions, review, and sign the assent and consent forms. The study was approved by the University of Cincinnati Institutional Review Board.

3.1.1. Introduction

Each session began with questions to assess children's and parents' awareness of wearables and the tracking capabilities available through these technologies. We also inquired about the child's existing technology ownership and whether any immediate family members owned wearables. This open-ended conversation served to ground the sessions in participants' lived experiences.

3.1.2. Wearables features

Next, children and parents individually defined the desired features of an ideal child-oriented wearable and were asked to arrange them in order of personal importance and preference. Parents were specifically asked to focus on what the wearable should track about their children, not themselves. They were not provided with a list of features to choose from, but they could see the options their child was given. Using sticky notes, they wrote features they wanted to track about their children and organized them.



Fig. 2. Hexagon blocks with features.

Children were given 24 preprinted hexagons, each representing a possible feature, along with 4 blank hexagons for custom additions. The hexagon shape was selected because it supports both hierarchical and circular arrangements, allowing children to spatially reflect the importance and relationships between features (Fig. 2). During study design, we initially tested octagons but found it difficult to represent hierarchical structures with them.

The features listed on hexagons were selected based on an analysis of commercial wearable devices. Five of the hexagons were not personal tracking metrics; they were utility-based (i.e., time, date, weather, timer, and stopwatch). The remaining hexagons focused on individual health and well-being features (i.e., active minutes, body temperature, calories burned, calorie intake, distance traveled, emotions, exercise, heart rate, location, medication, nutrition, respiration rate, sleep, stairs climbed, steps, stress, task reminders, water intake, weight). We intentionally excluded calling and texting features to see whether participants would independently identify these as desirable additions.

We also asked participants (both children and parents) to indicate who they wanted to give access to the data for each feature. They put stickers labeled with potential stakeholders (e.g., mom, dad, siblings) on their feature list. For calling and texting (if introduced by participants), stickers were used to specify who the child wanted to communicate with. We also asked clarifying questions about whether those contacts should have access to the call or message logs.

Before moving on to the wearable device design, we asked children whether the built-in screens of existing wearables would be sufficient for viewing information, or whether they would also like access to information on a larger external display, such as a tablet.

3.1.3. Wearable device

The wearable device design began with a discussion of the ideal location on the body. To support this, we provided a human figure cutout, prompting children to consider different placement options. Next, children were given materials to construct a physical model of the wearable they would like to own. These materials included cardstock, foam stickers, playdough, pre-cut screens and straps, colored markers, rulers, scissors, pencils, erasers, and glue.

Some children created detailed screen interface designs during this phase. For those who did not, we offered a hexagon screen cutout to help them describe what they would like to see first when viewing the screen.

While children were building their wearable models, parents engaged in parallel design activities. First, they were given smartphone screen printouts to illustrate what they would like to see in a companion app linked to their child's wearable. Following this, they received a wrist-worn wearable screen printout to indicate their preferences for what their child should see on the device's home screen.

3.1.4. Family use

The sessions concluded with a joint conversation about potential family-oriented uses of wearables, along with any additional comments

participants had regarding wearable devices, activity tracking, and data sharing. To support reflection during this discussion, we provided a weekly planner printout, prompting participants to consider their weekly routines and how a wearable might fit into their everyday lives.

3.2. Participants

We recruited participants using a combination of purposive and snowball sampling methods. As our focus was on the integration of wearables into everyday family life, we did not include any health-related inclusion criteria. Instead, we targeted neurotypical children aged 7 to 14, an age range chosen to capture elementary- and intermediate-school-aged children. This age range was determined by reviewing the target age ranges of commercially available wearables. We excluded high schoolers (children older than 14). Children differ significantly in their abstract thinking abilities, such as conceptualizing datafication, across ages (Alghythee et al., 2024; Kumar et al., 2017). Middle-to late-adolescent children often already own smartphones and typically exhibit greater independence, which may result in different perceptions of datafication and dataveillance.

All sessions were conducted in person, either at participants' homes, in participant-preferred public spaces (e.g., public libraries), in a university room, or at the first author's home. Session durations ranged from 39 to 60 min, with an average length of 48.8 min. Each child participant received a \$25 gift card at the conclusion of the session.

In total, we conducted 12 sessions with 12 parents and 15 children. In three of the sessions, multiple children from the same family participated. We ceased recruitment after 12 families (in addition to 5 pilot families) because no new codes emerged and existing themes were not modified, particularly those concerning wearable familiarity, value, child-parent conversations, and understanding of datafication and dataveillance.

Table 1 summarizes participants' ages and wearable ownership. The mean age of children was 9.7 years, with no child participant aged 14. The group included 10 daughters and 5 sons (C4.2, C6.1, C7, C10, and C12). All participating parents were mothers. Five of these mothers were not working at the time of the study, while the others were employed across a range of professions, including marketing and IT.

3.3. Data analysis

This paper reports findings from the main study involving 12 families and excludes data from the pilot study. All sessions were audio-recorded and subsequently transcribed using a transcription software. The verbal data were organized and coded in NVivo, while photographs documenting the artifacts created during the co-design activities were organized on a Miro board. Because our goal was to examine children's and parents' perspectives alongside the family interactions that occurred during the activities, we organized the data by family and treated the family as the unit of analysis. For each family, the transcripts and photographs were analyzed together, and the co-design artifacts were interpreted in relation to participants' verbal explanations and the interactions that occurred during their creation.

The first author conducted an inductive thematic analysis following (Braun & Clarke, 2006). The analysis involved familiarization with the interview transcripts and photographs, initial open coding, the iterative grouping of codes into preliminary themes, refinement of the themes with input from the research team, and subsequent interpretation and refinement through a socio-technical lens.

Initial open coding was conducted at the family level. It focused on children's and parents' familiarity with wearables, their expectations of wearables at both the feature and device levels, and their understanding of tracking and dataveillance. We also identified child-parent conversations and tensions that emerged during the co-design sessions and coded their topics and underlying reasons. Tensions were identified when differences in children's and parents' perspectives on wearable

Table 1
Participants' ages and wearable ownership. Children labeled as "used" did not personally own a wearable at the time of study but had used one before.

	F1		F2		F3		F4		F5		F6		F7		F8		F9		F10		F11		F12				
	C1	P1	C2	P2	C3.1	C3.2	P3	C4.1	C4.2	P4	C5	P5	C6.1	C6.2	P6	C7	P7	C8	P8	C9	P9	C10	P10	C11	P11	C12	P12
age	8		9		7	10		8	12		9		8	10		10		13		10		10		10		11	
wearable	Apple Air-Tag	Apple Watch	-	activity tracker (no brand information)	VTech	Apple Watch	Apple Watch	-	T-mobile SyncUP	-	-	-	VTech	Apple Watch	Apple Watch	-	-	used activity tracker	-	Gizmo	-	Gizmo	Apple Watch	used Garmin activity tracker & Gizmo	used activity tracker	used smartwatch	-
ownership																											



Fig. 3. C1's wearable and screen designs closely resemble her mother's Apple Watch.

features or data-sharing preferences prompted explanation, justification, disagreement, or resistance. We then compared children's and parents' perspectives within each family and examined codes across families for recurrence, variation, and contradiction. These comparisons guided the grouping of codes and the development of preliminary themes.

The themes captured participants' perspectives on the actual or envisioned benefits of wearables, the values attributed to their use, relational and family-level dynamics in data tracking and sharing, and the mechanisms shaping children's understanding of datafication and dataveillance. At this stage, we also gave special attention to instances in which child-parent conversations about data and datafication were absent or limited. Although explicit discussion of data collection and sharing was generally limited, some sessions included parent-led explanations or reflection on particular features. These contrasting cases led us to characterize such dialogue as episodic rather than entirely absent.

The research team brought expertise in personal and family informatics. These disciplinary perspectives informed our interpretation of family data practices and helped us situate the emerging themes within the existing literature. These discussions led us to refine the boundaries and labels of the core themes and clarify the relationships among them. During this later stage of the analysis, we also applied a socio-technical lens, specifically technological mediation (Verbeek, 2005). This lens guided our further examination of human-technology entanglement and interactions mediated through technology.

For clarity, in the remainder of the paper, we use the following abbreviations: F for family, C for child participant, and P for parent participant, followed by the assigned family number. In cases where a family included two child participants, we append 1 or 2 to the family number to differentiate between the children (e.g., C3.1, C3.2).

4. Findings

We present our findings as three analytical themes that explain how familiarity, values, and interactional dynamics shape children's encounters with wearables, datafication, and data sharing. Together, these themes reveal how data tracking becomes normalized through everyday use, often in the absence of sustained child-parent dialogue on datafication and data sharing.

4.1. Children's wearable familiarity and ownership without awareness of datafication

Children's wearable ownership was not novel for most participants. As shown in Table 1, only four of the 15 children had never owned or previously used a wearable device. Among these, two had family



Fig. 4. Examples of children's wearable screen designs (from left to right C4.2, C6.2, C8, and C9), illustrating similarities to existing commercial smart device interfaces.



Fig. 5. C6.1's smartphone and smartwatch are on the table, illustrating his existing ownership and use of smart technologies.

members who owned wearables, while the other two had seen such devices used by friends. Thus, all child participants were either directly or indirectly familiar with wearable technologies, including smartwatches and activity trackers.

For children who owned wearables, acquisition occurred either through parental decisions made independently of the child or through child-driven requests, often influenced by parents', siblings', or peers' wearable ownership, or advertisements. In some families (e.g., F10), wearables were passed down from older to younger siblings when the older sibling transitioned to smartphone ownership. In others, each child in the family owned their own device (e.g., F3).

Both children and parents generally perceived wearable ownership - once children reach a certain age (as discussed in the next section) - as ordinary and part of contemporary everyday life. This ownership pattern was also evident in interactions between children and parents. For example, during one session, we observed the following exchange. When the same child was asked to design a wearable for herself and describe what she would want to see on its home screen, she created a design closely resembling her mother's Apple Watch (Fig. 3).

C1: "Mom told me that I can have an Apple Watch and call her."

P1: "Yes, when you get a little older."

Such resemblance between children's wearable and screen designs and commercially available technologies was not an isolated

occurrence. Across sessions, children's wearable and screen designs closely resembled existing commercial devices (Fig. 4). Many screen designs featured black touchscreens, circular or rounded rectangular app icons, and familiar layouts similar to those on smartphones, tablets, and existing wearables. These patterns indicate children's familiarity with and deep integration of smart devices in their everyday environments.

Their familiarity with smart devices was further evidenced by their comments and references to other smart technologies they owned. Two participants (C8 and C12) owned smartphones, and one (C6.1) had access to a Wi-Fi-only smartphone for use at home. All children except C1 owned tablets. Without being prompted, several participants (C1, C3.2, C8, C9, C10, C11) mentioned using conversational agents such as Alexa and Siri to obtain information or assistance, for example, to check the weather and dress appropriately. These findings indicate that children's everyday lives are already embedded within a broader ecosystem of smart and data-driven technologies (see Fig. 5).

This level of familiarity with and ownership of smart devices appears to foster trust in technology, which manifested in two key ways in our data. First, children expressed a preference for wearable devices with a wide range of features, indicating no clear purpose or goal for owning a wearable. Almost all children selected more features than their parents, with an average of 15 features, nearly double that of their parents. Children were generally familiar with the labels for tracking features, sometimes to their parents' surprise. For example, when C3.1 was selecting features, her mother explained what water intake was, even though the child had not inquired, which led to a "I know" response from her child. Similarly, F4 asked whether C4.1 knew the difference between a timer and a stopwatch, and learned that she did.

Second, some children expected wearables to reveal otherwise invisible aspects of their lives. For instance, C6.2 expressed a desire "to know my emotions when I don't know," while C8 explained her preference for tracking calories burned over calories consumed by emphasizing that the former cannot be directly perceived:

"I didn't really put a lot of thought into that, but I feel like, because calories intake, I know what I'm eating, so and I can, like, feel to a certain amount, like, if I'm full and how much I ate, but calories burned, you can't really see it in front of you, like you can see the food that you eat. So I feel like those calories intake, if I just look at a few, then I know what I'm eating, but I don't know how much calories I'm burning throughout the day." (C8)

In some cases, children extended this expectation to learn about the invisible beyond self-awareness, imagining wearables as tools for self-regulation and external support. For example, C10 described how he could use wearable data to understand and manage back pain by

monitoring stress levels and taking appropriate action. Two other participants suggested sharing their invisible data with certain people so that those people can understand or support them:

P2: "Why do you think your teachers, xxx [sibling] and xxx [friend] need to know about sleep?"

C2: "Because if I'm a little extra cranky, I don't need to ..."

P2: "Explain yourself?"

C2: "Yeah."

I: "Why do you think the teacher should know about your stress and emotion?"

C9: "So if I get like, really stressed. Like, they can help me find a way to calm down."

While these expectations may indicate potential benefits of wearable technologies for children, our data also reveals children's limited understanding of datafication. Although children were familiar with feature labels, they did not consistently recognize these features as data that could be accessed, interpreted, and shared. This gap became particularly visible during the data-sharing exercise, when children were asked to specify who should have access to different types of data.

In several cases, this activity prompted moments of realization. For example, C3.2 wanted to track how well she sleeps, but she did not want her parents to have access to this data. When her mother suggested using heart rate data to determine whether C3.2 is anxious or sleeping well, C3.2 realized that different features could provide information about others. She reacted by saying "Oh damn!" and laughed. Similarly, two children (C9 and C10) who had already used wearables were unaware that their parents had occasionally tracked their location. Although they did not respond negatively to this parental tracking, they only learned this during the sessions:

C9: "You can see me moving?"

P9: "I can see the little dot of your location changing as you walk. You didn't know that?"

C9: "No."

P9: "Surprise."

C9: "Creepy"

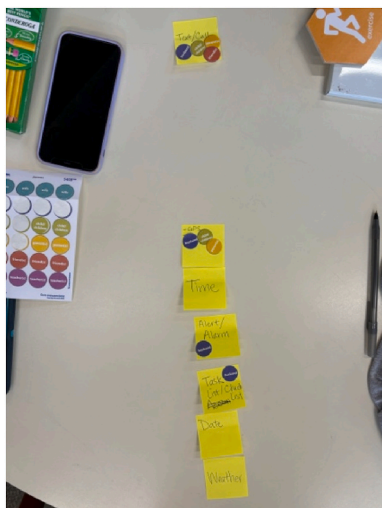


Fig. 6. P9's feature preference order highlights the prioritization of communication and access to the child.

P9: "Is it? I am your mama. Come on."

P10: "You have a watch like that, honey, that is your watch tracks your steps."

C10: "Yeah."

P10: "And tells the location where you are."

C10: "I do not know that far."

Overall, these findings suggest that children's familiarity with and ownership of wearable devices do not necessarily entail an understanding of how these technologies operate or how data is collected and shared. Current child-oriented wearables present data as bare features, are not transparent about datafication, and do not aim to teach children about it. More importantly, our sessions revealed an absence of explicit parental framing of datafication and data sharing, indicating that wearable adoption often occurs without deliberate child-parent conversation and negotiation of data practices. In this sense, current child-oriented wearables do not consistently support reflection; instead, their integration into everyday life often leaves underlying data practices uninterrogated while contributing to the normalization of data tracking and sharing at an early age without children understanding them. Moments in which children recognized these dynamics during our sessions often became points of tension and negotiation, which we explore in Section 4.3.

4.2. Wearables as a representation of family values and parental care

Parents' attitudes toward child-oriented wearables varied depending on the child's age. For younger children, parents often considered and purchased wearables as a type of "toy" (P3) and expressed limited interest in tracking features. However, as children grew older and began spending more time independently, parents increasingly valued wearables as devices for safety, communication, and assurance of well-being. This shift reflects a broader tension between supporting children's independence and maintaining parental oversight.

The perceived safety and independence enabled by wearables were primarily tied to continuous connectivity through calling, texting, and location tracking. This parental priority was also evident in their feature preferences. Parents selected an average of 8 features (range 4 to 13). *Location* (n = 12) was selected by all participants, followed by *text* (n = 10), *call* (n = 9), *task reminders* (n = 8), and *time* (n = 5). Notably, none of the parents cited health tracking as their primary motivation. This is unsurprising, as our child participants did not have chronic health conditions.

In general, parents viewed calling and texting as default functionalities for their child's wearable. This was clearly reflected in P9's feature preference order (Fig. 6). She ranked these features higher than others, emphasizing that the primary value of a wearable was to support her child's independence, such as walking around the neighborhood or walking to and from school, while maintaining mutual access when needed, particularly for safety reasons.

"I would put that way up high, because I think the main purpose of us having getting it for her was so that we could communicate back and forth as to where she was, what she needed, give her a call, tell her to come home, you know, like this basically fulfills all the purpose of her having a wearable for me ... So this is convenient. This is good information ... This is our main, our main priority, very high priority more than the others." (P9)

The other feature that parents valued and associated with safety was location tracking. Its importance was evident in both the parents' feature-preference orders and the home screen designs for the companion apps. All parents included location tracking on the home screens.

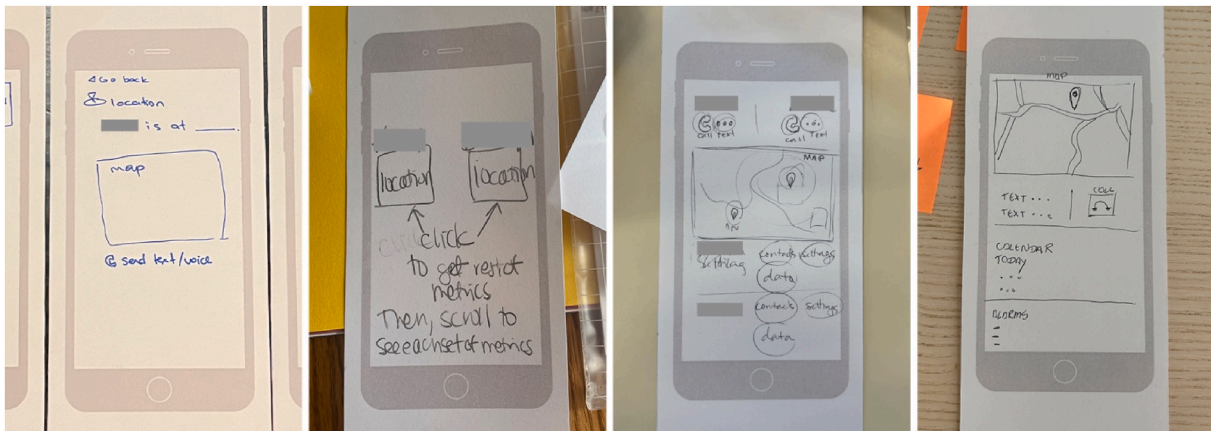


Fig. 7. Parent-designed app home screens (P1, P3, P9, and P10), illustrating the prioritization of location tracking on the home screen.

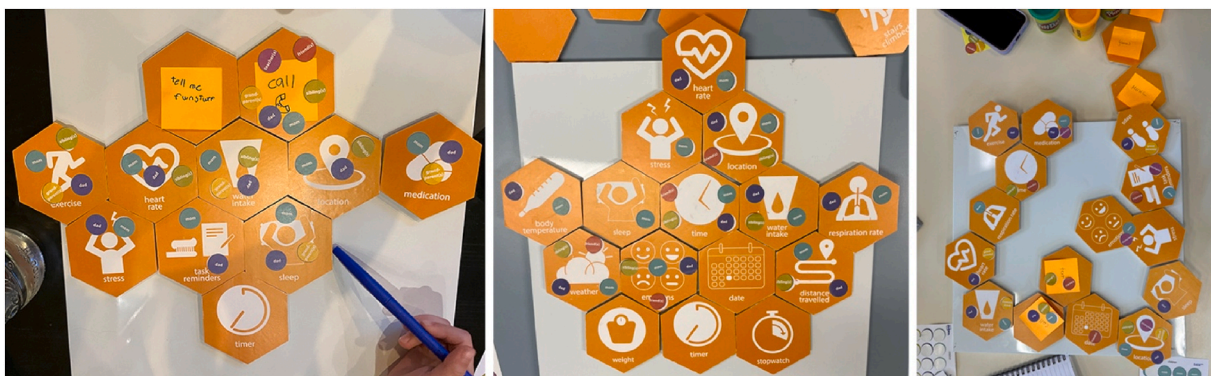


Fig. 8. Feature preferences, their order, and data-sharing preferences from C1, C7, and C9, illustrating both the range of desired tracking features and the willingness to share location data with parents and other people.

Eleven of the 12 parent participants placed it as the first feature in their app designs. Only F11 placed it second, with the call and text features listed first. In addition, five parents even made the location-tracking icon larger than the other icons on the screen (Fig. 7).

Like their parents, children also communicated the value of wearables for safety. Some children expressed feeling safer when their parents could track their location: “location is important because I could get lost or something” (C7). There were also children’s comments indicating that their parents had explicitly framed wearables as safety devices. For example, F1 purchased Apple AirTags for each child in the family to track their locations as they walked from school to aftercare programs supervised by school staff. C1 demonstrated a clear understanding of this purpose: “I place it on me and then my mommy can look at her phone, and then she can know where I am.”

Like C1, many participants considered location tracking to be essential. Notably, however, this importance was not always reflected in their feature orders (e.g., C9’s feature order in Fig. 8). Instead, they saw this feature as a family need and included it. Also, children who included location tracking consistently indicated a willingness to share this data with their parents (see Fig. 8). In these cases, location tracking was often valued more for parental reassurance than for the child’s own use.

Four children (C5, C6.1, C6.2, and C11) did not include location as a feature. However, among these four, three included call and text as features and placed stickers on them to indicate their desire to use these functions to connect with their parents and others (e.g., friends). For some children, connectivity via calls and texts was also framed as instrumental to safety and peace of mind. As C9 explained: “If I am apart from mom and dad, like, if I need anything or if I want anything, I can text them.”

The sense of safety provided by calling, texting, and location tracking was not the only value wearables offered. Some parents also expressed a desire to monitor aspects of their child’s life that are not easily observable. For instance, F4 commented that parents are typically in control of factors such as exercise, sleep, and nutrition, but not some other aspects of life: “I don’t know if they’re the ones who are controlling it. It’s really us who are controlling it. Whereas where they are, or what they are doing, they are in control of.” P8 selected stress as a feature to track in her daughter, explaining, “I wish she is happy and not, you know, stressed.” F2 maintained a structured sleep routine at home, but P2 still wanted confirmation that her children were following it:

“But I’m not sure how much they are sleeping, which can impact their following day, the day after. So even though lights out 8:30-9 o’clock, it may not be till 10 or 11 that they fall asleep. And they only get electronics on the weekends. So during the week, I don’t know what they’re doing to stay awake and be exhausted the next morning. So being able to at least track their sleep, because it’s such an important piece of good health.” (P2)

During our sessions, parents frequently compared wearables and smartphones, positioning wearables as alternatives to smartphones. They stated that wearables are unnecessary if the child already has a smartphone. However, all agreed that children should not own smartphones before a certain age, though this age varied across families. Wearables were viewed as a temporary solution that delayed smartphone ownership and enabled parents to track and communicate with their children through limited-function devices, particularly because of the lack of social media access. For example, P5 explained purchasing a wearable for her child if she could find one with limited connectivity: “I could get a device that had location and limited ability to text or call with my

phone. That would be something I would be interested in. So does not let her get into the internet and have all sorts of like other communications.”

Parents evaluated this limited connectivity as a wearable's unique value proposition. Some parents explicitly described wearables as “training tools for technology” (P9) and as transitional devices on the path to smartphone ownership. These transitional tools were seen as educational, helping children learn family values related to digital connectivity and accountability while supporting gradual technological socialization.

“I kind of see a watch as like, an intro to technology use, using so it would help me and my husband, kind of talk to her about how to use technology like a way, especially in these early times, of having technology to have family discussions about what we think is appropriate for technology and how to use it, so that by the time she does get a real phone or additional, more robust technology, that she would have some patterns and habits like worked out already ... we have a baseline of what our family expectations are and how to use things safely.” (P9)

These findings reveal how wearables function as representations of family values. Parents communicate and enact parental care by integrating child-oriented wearables into family life. Children, in turn, largely accept the value of this technology-mediated care practiced through parental tracking. In this sense, wearables become normalized as part of family life. Wearables mediate family dynamics through their role as mundane, taken-for-granted objects that become especially visible when child-parent connection is needed.

4.3. Wearables as mediators of child-parent dialogue and tension

While parents often described wearables as safety buffers for their children, they also expressed concerns about their use, particularly the potential for children to become obsessed with them. For example, F11 recalled a time when her older son had an activity tracker. He was required to remove it during soccer practices and occasionally asked his mother to shake the wearable during practice so he could receive credit for his physical activity.

Reflecting on such experiences and their own self-tracking, parents expressed apprehension about their children's engagement with self-tracking. They recognized the need to communicate these concerns to their children. Across many sessions, parents used co-design activities to initiate conversations, share perspectives, guide their children, and emphasize the importance of self-regulation over reliance on technology in managing health and well-being. For instance, C4.1 expressed a desire to track her weight using the wearable. Initially, P4 questioned whether C4.1 truly wanted the weight tracking feature. Later in the session, when the opportunity arose, P4 elaborated:

“I worry about having weight on these devices. I do not think children should worry about those kinds of things. You know what I mean, [C4.1].”

It can be kind of a very, I do not know, it is a hard one to know because as a child, there is no set weight you need to be either because you are growing. So, I feel worried that you have that on that one.” (P4)

We also observed moments when parents used opportunities to educate their children about specific features. This typically occurred when a child asked about an unfamiliar term. For example, when C5 expressed not knowing what a calorie meant, P5 explained:

“It is the energy that is inside food. So when you eat the food, it gives your body a certain amount of energy, which is measured as calories as a unit for that. And then when you exercise or do anything even think or whatever, it burns those calories and then you get hungry again eventually, right? Because you burned the calories you ate before. And so it's a constant cycle.” (P5)

These examples illustrate how structured co-design activities created space for child-parent discussions and revealed both children's and parents' openness to conversations about data and datafication. They also suggest the value of structured dialogue for identifying and addressing tensions that may otherwise remain unspoken.

The tensions we observed were particularly visible around data sharing. While children were generally comfortable with parents accessing location data, they were more selective about sharing other types of information. They mentally distinguish between features representing family values and their own personal values. For data they consider relevant to their personal values, they did not always grant access to both parents or even to a single parent. While parents emphasized data access for care and responsibility, children selected data recipients based on trust, anticipated reactions, and perceived judgment.

For example, C3.2 expressed disliking the idea of her parents tracking her screen time. At the time of the study, the family did not have a solution for tracking screen time across multiple devices (i.e., smartwatch, tablet, and TV) used by the children. When P3 expressed a desire for such a function, C3.2 questioned the need.

Similarly, C2 selected nutrition and weight as features in her wearable. She chose to share nutrition data only with her grandparents and weight data with her dad. When her mother asked her why she preferred to share nutrition data with grandparents, C2 explained: *“Because they are gonna be more gentle with it.”* Following the discussion of other selected features and C2's assignment of individuals to specific features, P2 reflected, *“what I noticed is ... her customization is [around] who won't judge her for these specific areas.”*

C8, the oldest child in our study, owned a smartphone that she already used to communicate with her parents. In addition to keeping in touch with her parents, she reported feeling comfortable sharing her location with them through a wearable device. Emotions were selected as a feature by both C8 and her parent during the study (Fig. 9). However, C8 did not want her parents to have access to her emotional data.

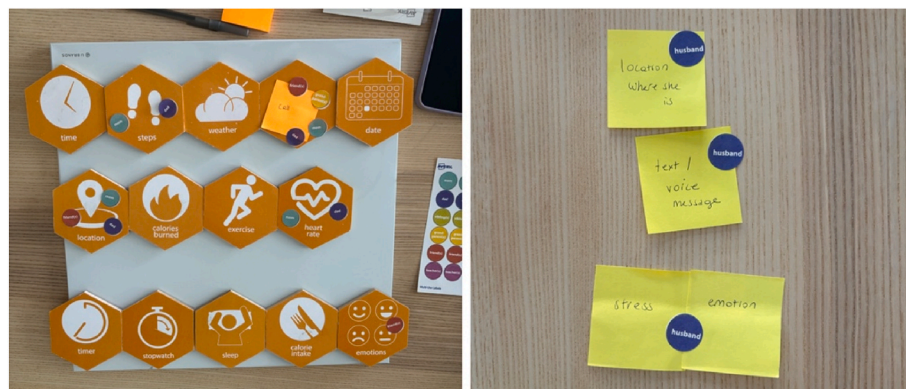


Fig. 9. C8's and P8's feature preferences and their differing data sharing preferences for emotions.

Instead, she preferred that her friends have access so they could support her when needed. She explained: *“And then for emotions, I did not put it because I feel, I think of emotions as kind of my privacy too. And then if they see that, I am maybe, like, angry at them or whatever, but I do not want it to be like a big thing, okay, whatever.”* While P8 expressed understanding of her daughter's perspective, she also emphasized the importance of knowing how C8 feels.

“She just said she does not want to share this emotion with parents, just friends, because she says privacy, but going through puberty ... sometimes there's something we can help as an adult. Friends can help as friends. It's different, so that's why I wanted to know. Sometimes she doesn't tell me. We know that something is going on and try to comfort her, so that is why I put this too.” (P8)

These interactions illustrate how wearables can function both as conversation starters and as sources of tension within families. While Section 4.1 highlighted the general absence of conversations about data practices, our findings here show that both parents and children are willing to engage in such discussions when given the opportunity. However, such discussions are not consistently present in everyday use; instead, they require mediation. The tensions observed underscore the necessity of these conversations, particularly in navigating differences between family values and children's emerging personal values.

5. Discussion

Through our co-design sessions, we observed complex family dynamics surrounding the use of child-oriented wearables, alongside the normalization of datafication and dataveillance, often without children fully understanding these concepts. In the following sections, we position our findings within the literature on the datafication of childhood, technology use in family life, and technological mediation, showing how they support, challenge, and extend existing work.

5.1. Family dynamics contributing to children's normalization of datafication and dataveillance

Findings from our co-design sessions support existing concerns in the literature about children's normalization of datafication and dataveillance without a full understanding of these concepts (Lupton & Williamson, 2017; Mascheroni, 2020) driven by the pervasive use of child-oriented wearables. More importantly, our data indicate that this normalization is not solely a consequence of exposure to these technologies at an early age. Rather, it is also shaped by the introduction of these technologies as tools of parental care that embody family values, as well as by the absence of sustained dialogue between parents and children about data, datafication, and dataveillance.

As parents adopt various forms of digital parenting technologies (Livingstone & Blum-Ross, 2020), they also use child-oriented wearables as parenting tools. Consistent with prior research (Hasinoff, 2017; Murphy & Wu, 2025; Oygur et al., 2021), parents primarily valued the peace of mind afforded by the ability to track and remain connected to their children without their children needing to own a smartphone. This way, parents ensure their children's safety and well-being. In this sense, our findings support existing work that frames such devices as expressions of parental care (Lupton, 2020; Sukk & Siibak, 2021). At the same time, we show that children are not passive recipients of these practices. They are aware of their parent's aim with the introduction of child-oriented wearables and interpret wearables as representations of family values and parental care.

Children's interpretation of wearables in this way suggests that these meanings are, at least implicitly, communicated within the family. However, our co-design sessions also revealed a notable absence of explicit child-parent dialogue about data, datafication, and dataveillance. While parents were willing to explain the meanings of specific features during co-design sessions, none attempted to explain how these

devices collect, quantify, and share data. Similarly, children rarely questioned these processes. Therefore, it was not surprising to see C9 and C10, two of the participants who owned wearables, react when their parents explained that they can see their location in real time in their apps.

Given prior research emphasizing the importance of child-parent interaction in shaping children's engagement with technology (Akter et al., 2022; Takeuchi & Stevens, 2011; Shin et al., 2021), this lack of dialogue is notable. In our case, it is not the presence but the absence of such conversations that appears to contribute to children's normalization of datafication and dataveillance without a full understanding of these concepts and processes.

These observations extend the current literature by identifying two interrelated factors that shape the normalization of data tracking and sharing without children's full comprehension of these concepts and processes. It is not only the introduction of child-oriented wearables into family life, but also their framing as expressions of family values and parental care, combined with the lack of sustained and explicit child-parent conversations about data and data practices, which appears to contribute to children's acceptance of these systems without questioning them or paying little attention to the underlying data practices.

Our study does not provide direct evidence explaining why parents avoid or defer these discussions. Consistent with Kumar et al. (2017) observations on online privacy, parents might be postponing such conversations, believing they will occur with more advanced smart device use, such as smartphones. But we also anticipate that parents might expect wearables themselves to play a role in teaching children about these concepts, as we explain in the next section.

5.2. Wearables as situational mediators of child-parent dialogue

Child-oriented wearables are associated with more complex use practices than their adult versions (Gram-Hansen, 2019; Jørgensen et al., 2016; Saksono et al., 2018). These practices involve interactions among children and wearables, parents and companion apps, parents and children, and occasionally among children and companion apps, as well as between parents and wearables. Analyzing these interrelated practices offers an opportunity to illustrate how technological mediation occurs through one specific co-used smart technology and to shift the focus from parents' role as mediators to technology's role in mediating family life.

Our data show that parents often view wearables as safer alternatives to smartphones and value them as tools to “train” (P9) children in responsible technology use. This framing suggests that parents implicitly expect wearables to mediate children's learning about data, responsible technology use, and digital behavior. However, as discussed earlier, children do not fully grasp the data practices underlying wearable features and often lack understanding of them. Taken together, these findings indicate that although the routine use of child-oriented wearables contributes to the normalization of datafication and dataveillance, current devices do not continuously support children's reflection on these processes. Instead, we observe situational and episodic forms of mediation, particularly in relation to child-parent dialogue about data and data sharing.

While in the previous section we discussed a lack of sustained dialogue between children and parents on data and data practices, this does not imply that dialogue is entirely absent. As noted in our findings, our co-design sessions included child-parent communication about these issues, with wearables alternately serving as conversation starters and sources of tension. Our sessions imply that this mediation occurs primarily in two contexts: (1) moments of structured interaction and (2) moments of conflict. As such, wearables can be understood as partial mediators of reflective family data conversations and negotiations.

In their role as conversation starters, wearables facilitated child-parent discussions about data, data tracking, and its implications on well-being. Parents used feature-selection activities during our sessions

to assess their children's familiarity with specific data types or to share perspectives and provide guidance. For example, P3 asked C3.1 whether she understood water intake, P5 explained the concept of calories, and P4 cautioned her daughter against tracking weight, citing concerns about potential obsessive behavior. These interactions illustrate how wearable features can act as entry points for dialogue, creating opportunities for learning and reflection through child-parent discussions. This aligns with prior work on co-use and joint media engagement, which highlights the importance of child-parent interaction in supporting children's digital literacy development (Akter et al., 2022; Takeuchi & Stevens, 2011).

At the same time, our findings show that wearables can generate moments of tension. Both children and parents framed wearables as tools of care. They are viewed as safety tools, enabled by location tracking and constant connectivity. Access to location data was rarely contested, perhaps because children regarded such data as “external”, consistent with (Czeskis et al., 2010), and therefore as more aligned with family-level information than with personal privacy. However, conflicts emerged around data that children perceived as more personal. For instance, C8 resisted sharing emotional data with her parents, preferring to share it with friends whom she believed could provide more appropriate support. Similarly, C2 selectively shared nutritional and weight-related data based on anticipated reaction and judgment. These findings resonate with prior research showing that children differentiate among types of data when sharing them (Czeskis et al., 2010; Thompson et al., 2024).

Taken together, these examples illustrate how children recognize and value parental care while simultaneously negotiating boundaries around personal data. Their decisions about data sharing are shaped not only by the type of data but also by anticipated reactions and perceived support from different recipients, highlighting a nuanced interplay between family values and children's emerging sense of autonomy.

These dynamics illustrate the situational nature of child-oriented wearables as mediators of reflection on data practices. While Peter-Paul Verbeek (2006) conceptualizes mediation as context-dependent, our findings extend this perspective by showing that mediation operates across multiple layers. Normalization occurs continuously through routine use, whereas reflection is situational and episodic, taking place during structured interaction or conflict.

5.3. Design implications to further benefit from child-oriented wearables as technological mediators

Our study suggests several design implications for enabling child-oriented wearables to support children's understanding of and reflection on datafication and dataveillance. Future designs should prioritize structured interactions for family co-use and reflection and increase transparency around datafication and dataveillance in ways that are accessible to children.

To support wearables in initiating child-parent dialogue on data and data practices - which our findings identified as critical to children's understanding of these concepts - designs should incorporate structured opportunities for interaction. For example, devices could prompt discussions about privacy, consent, and data sharing during initial setup and onboarding. At this stage, systems might engage both children and parents with questions that encourage conversation and help establish shared boundaries around data tracking and sharing. To foster sustained engagement, wearable interfaces could also introduce occasional, data-driven gamified tasks that require synchronized input from both the child and the parent, thereby creating recurring opportunities for dialogue.

In addition to facilitating dialogue, child-oriented wearables should be designed to enhance children's understanding of data practices through greater transparency and agency. Future designs could make data access more visible, interpretable, and meaningful to children. For example, wearable devices could notify children when their location

data is accessed through a linked app. More importantly, systems should enable children to make informed choices about which data is tracked and shared. Rather than relying on default tracking configurations, setup processes could encourage child-parent conversations about which metrics to monitor and why, reinforcing the value of intentional and selective tracking.

Beyond feature-level transparency, wearables should also make the underlying processes of datafication more legible. Interfaces could move away from presenting devices merely as collections of features and instead frame them as communicative systems that reveal how data are collected, processed, and shared. Such an approach would help children develop a more comprehensive understanding of datafication and dataveillance.

By fostering child-parent dialogue and supporting children's comprehension of data practices, child-oriented wearables could better realize their potential as tools not only for responsible technology use but also for cultivating critical awareness of datafication and dataveillance.

6. Limitations and future work

This study has several limitations, which also suggest directions for future research. First, our sample included age diversity within a single geography. All participating parents were mothers. These sample characteristics may have influenced the perspectives captured. Fathers' roles, broader caregiving networks, perspectives from other geographies, and age-related differences remain underexplored.

Second, differences in parenting patterns and technological inequities resulting from socio-economic disadvantages (Livingstone et al., 2015; Saksono et al., 2018) and cultural differences (Garg & Sengupta, 2019) are well documented. However, socio-economic status was not a criterion for recruitment in this work, and our study does not include families from low socio-economic backgrounds. Our findings, therefore, cannot be generalized to all families, particularly those experiencing digital exclusion.

Third, the study design focused on co-design rather than longitudinal use or ethnographic data collection in participants' homes. While this approach allowed us to capture perceptions, preferences, conversations, and negotiations, it did not reveal how practices might unfold in real life, evolve over time, or change as children age into new developmental stages. All these limitations indicate further areas of research.

7. Conclusion

This paper contributes to family-centered design by showing how the adoption of child-oriented wearables, together with family dynamics, continuously normalizes data tracking and sharing, whereas these technologies mediate children's reflection on and understanding of such practices only situationally and episodically. At times, these devices spark conversations that foster learning and care; at other times, they create tensions around privacy.

For design, these insights underscore the need for wearable-mediated child-parent interactions and age-appropriate design approaches that make data access transparent, prompt conversations during setup and continued use, and allow children to exercise agency over what is shared and with whom. Embedding these mechanisms could transform wearables from instruments of one-sided monitoring into tools that genuinely support critical awareness around data practices. Our study highlights the importance of moving beyond technical affordances and parental mediation toward examining child-oriented wearables as technological mediators in family life. This shift in perspective is essential for shaping child-oriented wearables that respect both children's rights and family co-use.

Selection and participation

All participating children demonstrated age-appropriate reading and writing literacy. Parents were asked to obtain their children's permission before scheduling the co-design sessions. At the start of each session, the study procedures and the children's roles were explained verbally. Children were provided with assent forms and given time to review them and ask questions. They were reminded that participation was voluntary and that they could withdraw at any time. The study received approval from the University of Cincinnati Institutional Review Board.

Declaration of AI Tools

The authors used ChatGPT during the preparation of this work for grammar checking, improving writing clarity, and copyediting. All AI-generated suggestions and revisions were reviewed and edited by the authors prior to inclusion in the published article.

CRedit authorship contribution statement

Isil Oygur: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yunan Chen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Daniel A. Epstein:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

References

- Akter, M., Godfrey, A. J., Kropczynski, J., Lipford, H. R., & Wisniewski, P. J. (2022). From parental control to joint family oversight: Can parents and teens manage mobile online safety and privacy as equals? *Proceedings of the ACM on Human-Computer Interaction*, 6(CSCW1), Article 57. <https://doi.org/10.1145/3512904>
- Alghythee, K. K. A., Hrnica, A., Singh, K., Kunisetty, S., Yao, Y., & Soni, N. (2024). Towards understanding family privacy and security literacy conversations at home: Design implications for privacy literacy interfaces. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)* (pp. 1–12). <https://doi.org/10.1145/3613904.3641962>, 983.
- Ali, S., Elgharabawy, M., Duchaussoy, Q., Mannan, M., & Youssef, A. (2020). Betrayed by the guardian: Security and privacy risks of parental control solutions. In *Proceedings of the 36th Annual Computer Security Applications Conference (ACSAC '20)* (pp. 69–83). <https://doi.org/10.1145/3427228.3427287>
- Barron, B., Martin, C. K., Takeuchi, L., & Fithian, R. (2009). Parents as learning partners in the development of technological fluency. *International Journal of Learning and Media*, 1(2), 55–77. <https://doi.org/10.1162/ijlm.2009.0021>
- Bettany, S. M., & Kerrane, B. (2016). The socio-materiality of parental style: Negotiating the multiple affordances of parenting and child welfare within the new child surveillance technology market. *European Journal of Marketing*, 50(11), 2041–2066. <https://doi.org/10.1108/EJM-07-2015-0437>
- Blackwell, L., Gardiner, E., & Schoenebeck, S. (2016). Managing expectations: Technology tensions among parents and teens. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW'16)* (pp. 1390–1401). <https://doi.org/10.1145/2818048.2819928>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- Cagiltay, B., Ho, H.-R., Michaelis, J. E., & Mutlu, B. (2020). Investigating family perceptions and design preferences for an in-home robot. In *Proceedings of the Interaction Design and Children Conference (IDC '20)* (pp. 229–242). <https://doi.org/10.1145/3392063.3394411>
- Cha, Y. J., Chen, J., Gunal, Y., Zhu, Q., Newman, M. W., & Park, S. Y. (2025). Collaborative health-tracking technologies for children and parents: A review of current studies and directions for future research. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)* (pp. 1–13). <https://doi.org/10.1145/3706598.3713596>, 1038.
- Cibrian, F. L., Lakes, K. D., Tavakoulia, A., Guzman, K., Schuck, S., & Hayes, G. R. (2020). Supporting self-regulation of children with ADHD using wearables: Tensions and design challenges. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)* (pp. 1–13). <https://doi.org/10.1145/3313831.3376837>
- Clark, L. S. (2011). Parental mediation theory for the digital age. *Communication Theory*, 21(4), 323–343. <https://doi.org/10.1111/j.1468-2885.2011.01391.x>
- Coyne, S. M., Padilla-Walker, L. M., & Howard, E. (2013). Emerging in a digital world: A decade review of media use, effects, and gratifications in emerging adulthood. *Emerging Adulthood*, 1(2), 125–137. <https://doi.org/10.1177/2167696813479782>
- Czeskis, A., Dermendjieva, I., Hussein, Y., Boring, A., Friedman, B., Gill, B., & Kohno, T. (2010). Parenting from the pocket: Value tensions and technical directions for secure and private parent-teen mobile safety. In *Proceedings of the Sixth Symposium on Usable Privacy and Security (SOUPS '10)* (pp. 1–15). <https://doi.org/10.1145/1837110.1837130>, 15.
- Ervasti, M., Laitakari, J., & Hillukkala, M. (2016). 'I want to know where my child is at all times' – Field study of a location-aware safety service for schoolchildren. *Behaviour & Information Technology*, 35(10), 833–852. <https://doi.org/10.1080/0144929X.2016.1201144>
- Fúster, J., Solera-Cotanilla, S., Pérez, J., Vega-Barbas, M., Palacios, R., Álvarez-Campana, M., & Lopez, G. (2024). Analysis of security and privacy issues in wearables for minors. *Wireless Networks*, 30(6), 5437–5453. <https://doi.org/10.1007/s11276-022-03211-6>
- Garg, R., & Sengupta, S. (2019). "When you can do it, why can't I?": Racial and socioeconomic differences in family technology use and non-use. In *Proceedings of the ACM on Human-Computer Interaction*, 3 pp. 1–22. <https://doi.org/10.1145/3359165>, 63, CSCW.
- Ghosh, A. K., Badillo-Urquiola, K., Guha, S., LaViola, J. J., Jr., & Wisniewski, P. J. (2018). Safety vs. surveillance: What children have to say about Mobile apps for parental control. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–14). <https://doi.org/10.1145/3173574.3173698>, 124.
- Global Market Insights. (2025). *Kids smartwatch market size*. Retrieved October 9, 2025 from <https://www.gminsights.com/industry-analysis/kids-smartwatch-market>.
- Gram-Hansen, S. B. (2021). Family wearables – What makes them persuasive? *Behaviour & Information Technology*, 40(4), 385–397. <https://doi.org/10.1080/0144929X.2019.1694993>, 2019.
- Hasinoff, A. A. (2017). Where are you? Location tracking and the promise of child safety. *Television & New Media*, 18(6), 496–512. <https://doi.org/10.1177/1527476416680450>
- Jennings, N. (2020). Family practices and digital technology. In L. Lievrouw, & B. D. Loader (Eds.), *Routledge handbook of digital media and communication* (pp. 154–165). Routledge.
- Jørgensen, M. S., Nissen, F. K., Paay, J., Kjeldskov, J., & Skov, M. B. (2016). Monitoring children's physical activity and sleep: A study of surveillance and information disclosure. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction* (pp. 50–58). <https://doi.org/10.1145/3010915.3010936>
- Kaziunas, E., Ackerman, M. S., Lindtner, S., & Lee, J. M. (2017). Caring through data: Attending to the social and emotional experiences of health datafication. In *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing (CSCW '17)* (pp. 2260–2272). <https://doi.org/10.1145/2998181.2998303>
- Koumpouras, Y., & Toulas, T. (2020). User centered design and assessment of a wearable application for children with autistic spectrum disorder supporting daily activities. In *Proceedings of the 13th ACM international conference on pervasive technologies related to assistive environments (PETRA '20)*, June 30, 2020 (pp. 1–9). <https://doi.org/10.1145/3389189.3398002>
- Kumar, P., Naik, S. M., Ramesh Devkar, U., Chetty, M., Clegg, T. L., & Vitak, J. (2017). "No telling passcodes out because they're private": Understanding children's mental models of privacy and security online. In *Proceedings of the ACM on Human-Computer Interaction*, 1 pp. 1–21. <https://doi.org/10.1145/3134699>, 64, CSCW.
- Kuzminykh, A., & Lank, E. (2019). How much is too much? Understanding the information needs of parents of young children. In *Proceedings ACM Interactive, Mobile, Wearable Ubiquitous Technologies*, 3 pp. 1–21. <https://doi.org/10.1145/3328923>, 52.
- Lee, H., Jung, Y., Shin, Y., Park, H., Choi, W., & Lee, U. (2024). FamilyScope: Visualizing affective aspects of family social interactions using passive sensor data. In *Proceedings of the ACM Human-Computer Interaction*, 8, Article 57. <https://doi.org/10.1145/3637334>, CSCW1.
- Liu, L., Zhang, C., & Lu, Z. (2024). Wrist-bound guanxi, jiazuo, and kuolie: Unpacking Chinese adolescent smartwatch-mediated socialization. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)* (pp. 1–21). <https://doi.org/10.1145/3613904.3642044>, 906.
- Livingstone, S. (2002). *Young people and new media: Childhood and the changing media environment* (1st ed. ed.). London; Thousand Oaks, Calif: SAGE.
- Livingstone, S., & Blum-Ross, A. (2020). *Parenting for a digital future: How hopes and fears about technology shape children's lives*. New York, NY: Oxford University Press.

- Livingstone, S., & Helsper, E. J. (2008). Parental mediation of children's internet use. *Journal of Broadcasting & Electronic Media*, 52(4), 581–599. <https://doi.org/10.1080/08838150802437396>
- Livingstone, S., Mascheroni, G., Dreier, M., Chaudron, S., & Lagae, K. (2015). *How parents of young children manage digital devices at home: The role of income, education and parental style*. London: EU Kids Online, LSE. Retrieved August 21, 2025 from <http://eprints.lse.ac.uk/63378>.
- Lupton, D. (2020). Caring dataveillance: Women's use of apps to monitor pregnancy and children. In L. Green, D. Holloway, K. Stevenson, L. Haddon, & T. Leaver (Eds.), *The Routledge Companion to Digital Media and Children* (pp. 393–402). Routledge.
- Lupton, D., & Williamson, B. (2017). The datafied child: The dataveillance of children and implications for their rights. *New Media & Society*, 19(5), 780–794. <https://doi.org/10.1177/1461444816686328>
- Mackintosh, K. A., Chappel, S. E., Salmon, J., Timperio, A., Ball, K., Brown, H., Macfarlane, S., & Ridgers, N. D. (2019). Parental perspectives of a wearable activity tracker for children younger than 13 years: Acceptability and usability study. *JMIR mHealth and uHealth*, 7(11), Article e13858. <https://doi.org/10.2196/13858>
- Manches, A., & Plowman, L. (2021). Smart toys and children's understanding of personal data. *International Journal of Child-Computer Interaction*, 30, Article 100333. <https://doi.org/10.1016/j.ijcci.2021.100333>
- Mancini, C., Rogers, Y., Thomas, K., Joinson, A. N., Price, B. A., Bandara, A. K., Jedrzejczyk, L., & Nuseibeh, B. (2011). In the best families: Tracking and relationships. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI'11)* (pp. 2419–2428). <https://doi.org/10.1145/1978942.1979296>
- Mascheroni, G. (2020). Datafied childhoods: Contextualising datafication in everyday life. *Current Sociology*, 68(6), 798–813. <https://doi.org/10.1177/0011392118807534>
- Mascheroni, G., & Siibak, A. (2021). *Datafied childhoods*. New York: Peter Lang.
- Mavoa, J., Simon, C., & Nansen, B. (2023). "It's about safety not snooping": Parental attitudes to child tracking technologies and geolocation data. *Surveillance and Society*, 21(1), 45–60. <https://doi.org/10.24908/ss.v21i1.15719>
- Murphy, R., & Wu, G. (2025). Why do rural migrant mothers in urban China digitally monitor their children? *Gender & Society*, 39(1), 91–119. <https://doi.org/10.1177/08912432241305605>
- Nouwen, M., Jafarainaimi, N., & Zaman, B. (2017). Parental controls: Reimagining technologies for parent-child interaction. *Proceedings of 15th European Conference on Computer-Supported Cooperative Work - Exploratory Papers*. (pp. 18–34). European Society for Socially Embedded Technologies (EUSSET). <https://doi.org/10.18420/ecscw2017-28>. PISSN: 2510-2591.
- Oygur, I., Epstein, D. A., & Chen, Y. (2020). Raising the responsible child: Collaborative work in the use of activity trackers for children. *Proceedings of the ACM on Human-Computer Interaction*, 4(CSCW2), 1–23. <https://doi.org/10.1145/3415228>, 157.
- Oygur, I., Su, Z., Epstein, D. A., & Chen, Y. (2021). The lived experience of child-owned wearables: Comparing children's and parents' perspectives on activity tracking. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI'21)* (pp. 1–12). <https://doi.org/10.1145/3411764.3445376>, 480.
- Oygur Ilhan, I., Chen, Y., & Epstein, D. A. (2024). Analyzing user experience with a smart product-service system: Children-owned wearables. In C. Gray, P. Hekkert, L. Forlano, & P. Ciuccarelli (Eds.), *DRS2024* (pp. 23–28). <https://doi.org/10.21606/drs.2024.403>. Boston, USA.
- Park, J. K., Akter, M., Wisniewski, P., & Badillo-Urquiola, K. (2024). It's still complicated: From privacy-invasive parental control to teen-centric solutions for digital resilience. *IEEE Security and Privacy*, 22(5), 52–62. <https://doi.org/10.1109/MSEC.2024.3417804>
- Patel, D., Chandola, V., & Bhat, S. (2024). *Kids smartwatch market report*. Dataintel. Retrieved October 9, 2025 from <https://dataintel.com/report/global-kids-smart-watch-market>.
- Pina, L. R., Sien, S.-W., Song, C., Ward, T. M., Fogarty, J., Munson, S. A., & Kientz, J. A. (2020). DreamCatcher: Exploring how parents and school-age children can track and review sleep information together. *PACM on Human-Computer Interaction*, 70, 1–25. <https://doi.org/10.1145/3392882>, 70.
- Pina, L. R., Sien, S.-W., Ward, T., Yip, J. C., Munson, S. A., Fogarty, J., & Kientz, J. A. (2017). From personal informatics to family informatics: Understanding family practices around health monitoring. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW'17)* (pp. 2300–2315). <https://doi.org/10.1145/2998181.2998362>
- Saatjohann, C., Ising, F., Krings, L., & Schinzel, S. (2020). STALK: Security analysis of smartwatches for kids. In *Proceedings of the 15th International Conference on Availability, Reliability and Security (ARES '20)* (pp. 1–10). <https://doi.org/10.1145/3407023.3407037>
- Saksono, H., Castaneda-Sceppa, C., Hoffman, J., El-Nasr, M. S., Morris, V., & Parker, A. G. (2018). Family health promotion in low-SES neighborhoods: A two-month study of wearable activity tracking. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)* (pp. 1–13). <https://doi.org/10.1145/3173574.3173883>
- Saksono, H., Castaneda-Sceppa, C., Hoffman, J., Morris, V., El-Nasr, M. S., & Parker, A. G. (2020). Storywell: Designing for family fitness app motivation by using social rewards and reflection. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)* (pp. 1–13). <https://doi.org/10.1145/3313831.3376686>
- Saksono, H., Ranade, A., Kamarthi, G., Castaneda-Sceppa, C., Hoffman, J. A., Wirth, C., & Parker, A. G. (2015). Spaceship launch: Designing a collaborative exergame for families. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW'15)* (pp. 1776–1787). <https://doi.org/10.1145/2675133.2675159>
- Schaeferbauer, C. L., Khan, D. U., Le, A., Garrett, S., & Siek, K. A. (2015). Snack buddy: Supporting healthy snacking in low socioeconomic status families. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)* (pp. 1045–1057). <https://doi.org/10.1145/2675133.2675180>
- Shin, J. Y., & Holtz, B. (2020). Identifying opportunities and challenges: How children use technologies for managing diabetes. In *Proceedings of the Interaction Design and Children Conference (IDC '20)* (pp. 495–507). <https://doi.org/10.1145/3392063.3394444>
- Shin, J. Y., Rheu, M., Huh-Yoo, J., & Peng, W. (2021). Designing technologies to support parent-child relationships: A review of current findings and suggestions for future directions. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2), 1–31. <https://doi.org/10.1145/3479585>, 441.
- Silva, L. M., Cibrán, F. L., Bonang, C., Bhattacharya, A., Min, A., Monteiro, E. M., Armando Beltran, J., Schuck, S., Lakes, K. D., Hayes, G. R., & Epstein, D. A. (2024). Co-Designing situated displays for family co-regulation with ADHD children. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)* (pp. 1–19). <https://doi.org/10.1145/3613904.3642745>, 124.
- Silva, L. M., Cibrán, F. L., Monteiro, E., Bhattacharya, A., Beltran, J. A., Bonang, C., Epstein, D. A., Schuck, S. E. B., Lakes, K. D., & Hayes, G. R. (2023). Unpacking the lived experiences of smartwatch mediated self and co-regulation with ADHD children. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)* (pp. 1–19). <https://doi.org/10.1145/3544548.3581316>, 90.
- Simpson, B. (2014). Tracking children, constructing fear: GPS and the manufacture of family safety. *Information and Communications Technology Law*, 23(3), 273–285. <https://doi.org/10.1080/13600834.2014.970377>
- Su, Z., & Chen, Y. (2024). Data-driven technology for children's health and wellbeing: A systematic review. *Foundations and Trends in Human-Computer Interaction*, 18(1), 1–99. <https://doi.org/10.1561/1100000090>
- Sukk, M., & Siibak, A. (2021). Caring dataveillance and the construction of "good parenting": Estonian parents' and pre-teens' reflections on the use of tracking technologies. *Communications: The European Journal of Communication Research*, 46(3), 446. <https://doi.org/10.1515/commun-2021-0045>
- Sun, K., Kanchi, R., Ello, F. M. T., Co, L.-N., Wu, M., Gelman, S. A., Radesky, J., Schaub, F., & Yip, J. (2024). "Why is everything in the cloud?": Co-designing visual cues representing data processes with children. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference (IDC '24)* (pp. 517–532). <https://doi.org/10.1145/3628516.3655819>
- Takeuchi, L., & Stevens, R. (2011). *The new coviewing: Designing for learning through joint media engagement*. New York: The Joan Ganz Cooney Center at Sesame Workshop. Retrieved December 29, 2020 from <https://joanganzcooneycenter.org/publication/the-new-covieing-designing-for-learning-through-joint-media-engagement/>.
- Thompson, L., Sydney, C., Beaumont, K., Brigden, A., & Bird, J. (2024). Exploring young children's views on sharing personal health data between ages 7–14. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference (IDC '24)* (pp. 800–805). <https://doi.org/10.1145/3628516.3659397>
- van der Kamp, M. R., Klaver, E. C., Thio, B. J., Driessen, J. M. M., de Jongh, F. H. C., Tabak, M., van der Palen, J., & Hermens, H. J. (2020). WEARCON: Wearable home monitoring in children with asthma reveals a strong association with hospital based assessment of asthma control. *BMC Med Inform Decis Mak*, 20, Article 192. <https://doi.org/10.1186/s12911-020-01210-1>
- Vasalou, A., Oostveen, A.-M., & Joinson, A. N. (2012). A case study of non-adoption: The values of location tracking in the family. In *Proceedings of the ACM 2012 conference on Computer Supported Cooperative Work (CSCW '12)* (pp. 779–788). <https://doi.org/10.1145/2145204.2145321>
- Vaterlaus, J. M., Beckert, T. E., & Bird, C. V. (2015). "At a certain age it's not appropriate to monitor one's child": Perceptions of parental mediation of emerging adult interactive technology use. *Emerging Adulthood*, 3(5), 353–358. <https://doi.org/10.1177/2167696815581277>
- Verbeek, P.-P. (2005). *What things do: Philosophical reflections on technology, agency, and design*. University Park, PA: Penn State University Press.
- Verbeek, P.-P. (2006). Materializing morality: Design ethics and technological mediation. *Science, Technology & Human Values*, 31(3), 361–380. <https://doi.org/10.1177/0162243905285847>
- Verbeek, P.-P. (2016). Toward a theory of technological mediation: A program for postphenomenological research. In J. K. B. O. Friis, & R. P. Crease (Eds.), *Technoscience and postphenomenology: The manhattan papers* (pp. 189–204). London: Lexington Books.
- Wang, J., Ann O'Kane, A., Newhouse, N., Sethu-Jones, G. R., & Barbaro, K. de (2017). Quantified baby: Parenting and the use of a baby wearable in the wild. *Proceedings of the ACM on Human-Computer Interaction*, 1(CSCW), 1–19. <https://doi.org/10.1145/3134743>, 108.
- Yip, J. C., Clegg, T., Ahn, J., Odili Uchidiuno, J., Bonsignore, E., Beck, A., Pauw, D., & Kelly, M. (2016). The evolution of engagements and social bonds during child-parent co-design. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)* (pp. 3607–3619). <https://doi.org/10.1145/2858036.2858380>
- Zehring, R., Huang, L., Lee, B., & Kyoung Choe, E. (2021). Investigating opportunities to support kids' agency and well-being: A review of kids' wearables. *arXiv*. Retrieved July 6, 2021 from <http://arxiv.org/abs/2104.05979>.